Reconstruction Fidelity and Perplexity Reliability in Afrikaans Latent Text Diffusion

Hadi Butt
School of Computer Science and Applied Mathematics
University of the Witwatersrand, Johannesburg

Abstract

Latent diffusion generates text by denoising in a continuous space and decoding the result, so every sample passes through the autoencoder that defines that space. We examine what this means when the pipeline is adapted to Afrikaans with limited in-domain data (NCHLT), using an mT5-base encoder–decoder with a Perceiver bottleneck and two diffusion denoisers trained with five seeds each. On 81 validation passages the original autoencoder reproduces 11 exactly and keeps 21 of 33 numerals; shuffling latents across documents shows that the decoder uses the latent, not that the latent preserves the passage. Generation scores do not reveal this. On test samples the larger denoiser lowers external mGPT perplexity from 51.31 to 48.61, yet on validation 1,000 random latents decoded by the same decoder reach similar raw perplexity although only 5.5% of their outputs are identified as Afrikaans, a second Afrikaans scorer prefers them to diffusion output, and a Gaussian fitted to real latents yields Afrikaans text that the larger denoiser beats only narrowly. A revised training recipe that trains every autoencoder parameter reconstructs 73 of 103 reserve passages exactly, against 19 for the original; its effect on generation is not yet measured. Reconstruction, language identity and generation scores measure different properties and should be reported together.

1 Introduction

A government notice states that the Departement van Maatskaplike Dienste en Armoedeverligting (Social Services and Poverty Alleviation) offers 44 bursaries a year, 20 of them for social development. Encoded into a latent and decoded again by an Afrikaans text autoencoder, the passage keeps 44, loses 20, and the department becomes "Maatskaplike Ontwikkeling" (Social Development). Latent text diffusion generates through such an autoencoder: a denoiser produces a latent, and a decoder turns it into words (Lovelace et al., 2023). In unconditional generation there is no source passage to compare against, so scores on generated text cannot show whether the autoencoder preserves content. Reconstruction does not measure the quality of generated text; it tests whether the latent space represents passages faithfully, which a latent generator relies on to express specific content such as names and numbers.

Latent text diffusion has been evaluated mainly on English, by scoring generated text (Li et al., 2022; Gong et al., 2023; Lovelace et al., 2023). Lovelace et al. (2023) report near-perfect reconstruction for English and, with a jointly fine-tuned mT5, for German, both with large training corpora. We found no study that tests whether such an autoencoder still preserves passages when adapted with limited in-domain data, or what generation scores show when it does not. Afrikaans is an informative case: the NCHLT (National Centre for Human Language Technology) corpus offers licensed government-domain text (Eiselen and Puttkammer, 2014) but only about 1.2 million training words, while Afrikaans is covered by a pretrained encoder–decoder, mT5 (Xue et al., 2021), and an external scorer, mGPT (Shliazhko et al., 2024). We ask:

1. **RQ1:** What constraints does a reduced English reproduction of the pipeline identify before the Afrikaans results are interpreted?
2. **RQ2:** Do autoencoders adapted to NCHLT Afrikaans preserve enough of a passage for faithful reconstruction?
3. **RQ3:** Does Afrikaans latent diffusion compare credibly with an autoregressive (AR) baseline under mGPT perplexity, diversity and repetition diagnostics, and tokenisation fertility?

Under criteria fixed in the research proposal before the experiments, the system is credible if its reconstructions preserve content, its mGPT perplexity is at most 1.20 times that of the AR baseline, and its outputs show no severe repetition or loss of diversity. The AR baseline shares corpus and tokeniser but is trained from scratch and sampled with nucleus sampling, whereas the diffusion system decodes with a pretrained decoder and beam search, so the comparison sets two systems against each other, not two generation mechanisms.

We answer these proposal questions mainly as a diagnostic study. Our central finding is that favourable generation scores and unfaithful reconstruction coexist in the same system, so raw perplexity did not establish reconstruction fidelity. We contribute (i) a reconstruction analysis that separates use of the latent from preservation of the passage; (ii) evidence, from random and fitted-Gaussian latent controls, language identification and a second scorer, that raw generation scores do not track this failure and can favour non-Afrikaans output; (iii) a follow-up autoencoder screen with rules fixed in advance, showing that a revised training recipe largely removes the reconstruction deficit at the original latent size; and (iv) the same checks applied to a Swahili run, showing which parts of the pattern recur.

2 Related work

Diffusion models generate by reversing a gradual noising process (Ho et al., 2020; Nichol and Dhariwal, 2021). For text, tokens have been handled by diffusing over discrete states (Austin et al., 2021), over continuous embeddings (Li et al., 2022; Gong et al., 2023; Strudel et al., 2022; Dieleman et al., 2022), or with likelihood-trained continuous models (Gulrajani and Hashimoto, 2023). Latent diffusion instead denoises in an autoencoder's latent space (Rombach et al., 2022); Lovelace et al. (2023) apply this to language by compressing a pretrained encoder–decoder, and Zhang et al. (2023) extend it to paragraphs. A latent model can generate only what its decoder can express: strong decoders may ignore their latent (Bowman et al., 2016; Li et al., 2020), and bottleneck autoencoders trade compression against reconstruction (Montero et al., 2021).

Two recent studies also ask where quality is lost. Gavrilov et al. (2026) separate a codec's reconstruction loss from its latent generator's and find most loss before generation begins; Jiang et al. (2026) find that alignment of the latent space with a pretrained language model matters more for generation than reconstruction fidelity alone. Both use large English corpora; we study an autoencoder adapted with limited in-domain data, with controls showing what external scores reward when its reconstructions are unfaithful.

Open-ended generation has no reference output, so it is usually scored with an external language model and diversity statistics. Probability and quality can disagree (Holtzman et al., 2020; Meister et al., 2020), which motivated distribution-level measures (Pillutla et al., 2021), and any metric needs validating for its use (Reiter, 2018). Resources for African languages remain uneven (Joshi et al., 2020; Nekoto et al., 2020), although models pretrained on modest data can be competitive (Ogueji et al., 2021; Adebara et al., 2023) and Afrikaans-specific encoders exist (Ralethe, 2020). We found no evaluation of diffusion-based generation for Afrikaans.

3 Data and method

Corpus. We use the clean 2.0 distribution of NCHLT Afrikaans (government-domain text, CC BY 2.5 South Africa) (Eiselen and Puttkammer, 2014). Text is NFC- and whitespace-normalised, contact details are masked, and passages of at most 50 words are formed; passages under three words and exact duplicates are removed, and no language-identification filter is applied. Documents are split with seed 42 into 47,767 training, 5,905 validation and 5,692 test passages (2,005, 251 and 251 documents); no passage is shared across splits, although some 12-word sequences are: 168 of the 4,827 validation passages outside the reserve (64 of 187 documents) share at least one with training, the longest shared span has a median of 18 words (maximum 46), and only 2.4% of shared 12-grams occur in five or more training documents. We keep the mT5 tokeniser (250,100 entries, 128-token cap) instead of training a SentencePiece vocabulary (Kudo and Richardson, 2018) as proposed, because the pretrained embeddings depend on it; validation fertility is 1.933 tokens per word, and the cap truncates 95 training passages (0.20%) and 2 of the 4,827 validation passages.

Autoencoder and diffusion. Following Lovelace et al. (2023), the mT5-base encoder feeds a three-layer Perceiver (Jaegle et al., 2021) that compresses

its states to a latent of 32 slots (vectors) of dimension 64, each rescaled to norm eight, and a reconstruction map passes it to the mT5 decoder. Only the Perceiver components, the final decoder block and the final layer norm are trained, with teacher-forced cross-entropy (CE); we use update 30,500 (effective batch eight) of a run that continued an earlier pilot. The follow-up screen (§5.4) trains every parameter instead. Denoisers are trained to predict the velocity (Salimans and Ho, 2022) under a cosine schedule (Nichol and Dhariwal, 2021), with self-conditioning (Chen et al., 2023); Appendix A gives the training objective, sampler and scoring estimators as implemented. The code is a fork of the official implementation of Lovelace et al. (2023).

Systems. The small and large denoisers (width/depth/heads 256/4/4 and 768/12/12; 8.1M and 188M parameters) are trained with seeds 42–46, selecting the checkpoint with the lowest velocity mean squared error of an exponential moving average (EMA) of the weights on a validation probe. Each run samples latents with 250 denoising diffusion probabilistic model (DDPM) steps (Ho et al., 2020) and decodes them with beam search (four beams, repetition penalty 1.2, no repeated trigrams), producing 1,000 validation and 1,000 test samples. The AR baseline is a four-layer GPT-style model (width 256; 67M parameters, of which 64M, or 95%, are its tied 250,100-entry embedding) trained from scratch on the same splits (Table 1); its checkpoint at update 60,000 is selected on validation CE, and it produces 256 validation samples with nucleus sampling ($p = 0.95$, temperature one). (Beam search from a fixed start token would return one output.) Two latent controls replace the denoiser and are decoded with the same decoder (validation only, 1,000 latents each): isotropic Gaussian latents with each slot rescaled to norm eight, and draws from a Gaussian fitted to the 47,767 training latents (Ledoit–Wolf shrinkage), projected to norm eight per slot.

4 Evaluation

Reconstruction. On 81 validation passages from 64 documents we report teacher-forced CE and, for free-running output, exact matches against the 128-token model view, mean sentence chrF (Popović, 2015), and numeral recall and precision, counting numerals (`\d+([.,]\d+)*`) as multisets, so dropped, duplicated and invented numerals all count, though not which entity a number belongs to. As a control, each latent is replaced by that of a passage from another document of similar length. An AI assistant that saw each source beside its reconstruction also labelled all 81 against three pre-specified content checks (entities and content words preserved, main clause recognisable, nothing unsupported); no Afrikaans reader verified the labels.

Generation. All text is scored with mGPT (FP32, 512-token window, stride 256), each output separately after a start token. Perplexity (PPL) is token-weighted over a sample set and is not the generator's own likelihood. We also report mean length, distinct-2, exact training copies and the share of outputs identified as Afrikaans by GlotLID v3 (Kargaran et al., 2023), with a dictionary check against the LibreOffice `af_ZA` Hunspell list. Because mGPT is strongly length-dependent, comparisons are repeated with length-standardised perplexity, which reweights per-band log-loss (1–10, 11–20, 21–35 and 36+ words) to the human length distribution, and under a second scorer, the monolingual Afrikaans Goldfish model (Chang et al., 2024). Goldfish's sources include Glot500, which contains NCHLT corpora, so overlap with our data cannot be ruled out. Evaluation sets differ in split and size (256 or 1,000 validation outputs, 1,000 test outputs per run); Appendix B lists every set, how it was chosen and when it was first used, and comparisons are made only within a set.

Uncertainty and data use. Diffusion values are means over five runs. The large-minus-small difference has a run bootstrap, a Welch interval and an exact permutation test over all 252 divisions of the ten runs; AR-ratio intervals omit AR and autoencoder training variation; reconstruction intervals resample documents. All intervals are exploratory. Checkpoints were selected on validation data, but the test set was exposed during the project: the autoencoder's test reconstruction metrics were reported on 19 September, and on 9 October the diffusion test perplexities were reported and an AI assistant rated 20 test outputs per configuration (unblinded, not reported here). From 23 October test banks were treated as an exposed benchmark with models frozen, so test results are not fresh confirmatory evidence. To test whether the checks matter beyond Afrikaans, we apply them unchanged to an earlier Swahili run of the pipeline (§5.5).

Condition	Small diffusion	Large diffusion	AR baseline
Pretrained weights	Adapted mT5	Same adapted mT5	None (random initialisation)
Generator parameters	8.10M	187.98M	67.22M
Training seeds	42–46	42–46	42
Precision / effective batch	FP32 / up to 64	FP32 / up to 64	BF16 / 8
Exposure at selection	288–306 epochs	172–180 epochs	480,000 sampled passages
Recorded training time	2.93–3.13 h	7.19–7.85 h	4.52 h
Text decoding	Beam 4, constrained	Beam 4, constrained	Nucleus, $p = 0.95$, temperature 1

Table 1: Comparison conditions (RQ3). All systems use the same splits and tokeniser. Diffusion parameter counts exclude the shared autoencoder; the AR count includes its tied 250,100-entry embedding, so counts are not whole-system equivalents. Training times cover training and checkpoint selection on an RTX PRO 6000 Blackwell and exclude pretraining and autoencoder training. AR passages are sampled with replacement, so they are not complete epochs.

Measure	Correct	Shuffled
Teacher-forced CE	1.079	8.319
Mean sentence chrF	67.70	12.84
Exact reconstructions	11/81	0/81
Numeral recall	21/33 (0.64)	0/33 (0.00)
Numeral precision	21/25 (0.84)	0/25 (0.00)
All three content checks (AI-assisted)	26/81	–

Table 2: Reconstruction of 81 validation passages from correct or shuffled latents (RQ2). Numerals are counted as multisets: 33 in the sources, 25 in each arm's outputs.

5 Results

5.1 Reconstruction

With correct latents teacher-forced CE is 1.079; with shuffled latents it is 8.319 (difference 7.240, 95% CI [6.605, 7.933]), and chrF falls from 67.70 to 12.84 (Table 2). The decoder therefore relies on source-specific information in the latent, yet the encode–decode path loses content: only 11 of 81 passages are reconstructed exactly and 21 of 33 numerals survive, while 4 numerals appear that are not in the source, with no input truncated and no output at the length limit. Earlier 1,024-passage evaluations agree (token accuracy 82.75%, exact reconstruction 12.21% on validation). Exact-match failure is not the same as a change of meaning, since a punctuation difference also fails it, but many errors do change content: "regulering" (regulation) becomes "betaling" (payment), and "Regeringsdepartement" (government department) becomes "Regeringskommissie" (commission). How often this happens has only model estimates. An AI labeller that saw each source beside its reconstruction marked 55 of 81 passages (68%) as failing at least one content check, and 62 (77%) under a stricter rule for seven borderline passages. These are unvalidated model labels, not measured error rates; neither was checked by an Afrikaans reader.

5.2 Generation

Denoiser size. The larger denoiser lowers test perplexity from 51.31 to 48.61 (Table 3; difference -2.704; Welch 95% CI $[-4.567, -0.841]$; exact permutation $p = 6/252 = 0.024$). Width, depth and training exposure change together, so this is the effect of the larger configuration as a whole. Its outputs are shorter and copy a training passage more often (79 against 5 of 5,000 test outputs), but removing copies leaves perplexity almost unchanged (validation, $5 \times 1{,}000$ outputs: 49.60 to 49.56). Exact copies understate the difference: on validation, 2.4% of the large configuration's word 8-grams occur in training against 0.4% for the small one (human text: 3.4%), and 12 to 20 of its outputs per run are near-copies (at least 90% of 8-grams shared), against 0 to 4.

AR comparison and the decoder alone. On 256 validation samples the raw ratios to AR are 0.6458 and 0.6087, well within 1.20. Random latents decoded by the same decoder, however, score 57.60 on the same 256 rows; on 1,000 validation latents they score 50.36, against 51.94 and 49.60 for the $5 \times 1{,}000$ validation outputs of small and large diffusion, so mGPT perplexity alone does not show what the denoiser adds. Human test text scores 33.62, but the scorer is strongly length-dependent: human passages of 1–10 words score 240.83 and those of 36+ words 25.93. Both diffusion sizes have higher perplexity than human text in every band longer than ten words (Figure 1). Resampling the AR model under 18 temperature, top-p and constraint settings found no better-matched comparator: every setting produced 36 to 112 exact training copies per 1,000 validation outputs, against at most

System	Validation, 256 samples		Test, 1,000 samples per run				Validation
	PPL	Ratio to AR	PPL	Mean words	Distinct-2	Train copies	Afrikaans
Human text	32.82	–	33.62	27.66	0.6796	0/1,000	98.2%
Random latents, same decoder	57.60	0.7009	–	–	–	–	5.5%
AR, nucleus sampling	82.18	–	–	–	–	–	95.3%
Small diffusion (5 runs)	53.07	0.6458	51.31 ± 1.49	29.30	0.6606	5/5,000	96.3%
Large diffusion (5 runs)	50.02	0.6087	48.61 ± 0.91	26.34	0.6903	79/5,000	97.2%

Table 3: Generation results (RQ3). Diffusion values are means over five runs (± sample SD). Ratio 95% intervals: small [0.5813, 0.7148], large [0.5477, 0.6736]. AR and random latents were evaluated on validation only. Afrikaans: share of outputs whose top GlotLID label is `afr_Latn`, on validation outputs (human: 4,825 passages, 95% interval 96.5–98.7%).

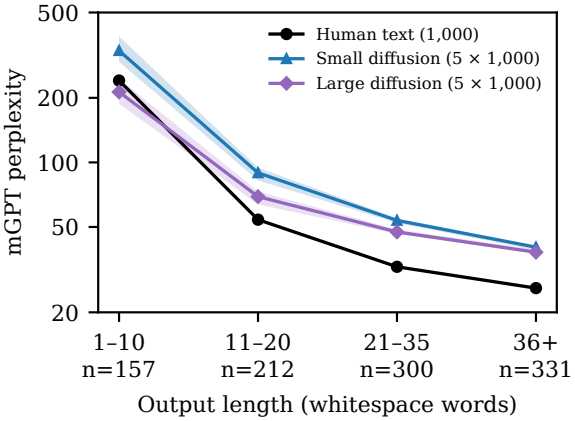


Figure 1: mGPT perplexity by output length on the test split, log scale: 1,000 human passages (counts per band under each label) and five runs of 1,000 outputs per diffusion size (line: mean of run values; shading: range). AR is omitted because it was evaluated on validation only.

Bank	Afrikaans	Unrec. tokens	4-gram overlap	Std. PPL mGPT	Std. PPL Goldfish
Human	98.2	4.3	14.4	–	–
Human, scored subset	97.9	4.2	15.7	41.6	64.6
Small diffusion	96.3	5.8	8.7	71.3	194.3
Large diffusion	97.2	5.9	11.8	59.6	148.1
AR, nucleus	95.3	5.3	15.8	80.5	203.7
Random latents	5.5	70.5	0.0	107.6	98.4
Fitted-Gaussian latents	99.7	4.6	8.3	63.6	172.5

Table 4: Validation outputs (%, except Std. PPL, the length-standardised perplexity). The 4,825-passage human pool was not scored by mGPT or Goldfish; scorer values are for an 811-passage human subset. Rows: 4,825 human passages; 5×1,000 per diffusion size; 256 AR; 1,000 latents per control. Unrec.: word tokens not in the `af_ZA` dictionary. 4-gram overlap: share of word 4-grams found in the training split. Human intervals (document bootstrap): Afrikaans 96.5–98.7, unrecognised 3.2–5.7, overlap 10.2–19.4.

13 for any diffusion run, so none met the pre-set eligibility rule. The AR baseline is thus a bounded from-scratch comparator, not evidence that diffusion outperforms autoregression.

5.3 Language, shared phrasing and a second scorer

Most diffusion outputs are identified as Afrikaans, at rates near human text (Table 4); the small configuration falls just below the human interval, mostly in outputs of ten words or fewer. Unrecognised words are slightly more frequent than in human text. In an AI-assisted reading of 50 unrecognised words per group, 22 from diffusion were malformed against 8 from human text. Diffusion outputs share 8.7% (small) and 11.8% (large) of their word 4-grams with the training split, against 14.4% for held-out human text. These checks describe the generated text; they do not link its errors causally to reconstruction errors.

The random-latent control shows why these checks matter. Its raw mGPT perplexity is close to diffusion's, yet only 5.5% of its outputs are identified as Afrikaans. Length-standardised mGPT ranks it last, but Goldfish does not penalise it: its raw Goldfish perplexity is 56.1 against 159.4 and 133.5 for small and large diffusion, and even after standardisation Goldfish ranks it best among the generated outputs, though still behind human text (Table 4). Per output, the two scorers' losses per token correlate only moderately (Spearman 0.54 to 0.73 for diffusion, AR and human text), and Goldfish rates 31% of the random outputs better than the median human passage, against 5% to 11% of diffusion outputs. Goldfish is therefore not a validated quality scorer for this use.

The fitted-Gaussian control asks a sharper question: does the denoiser beat a simple model of where real latents lie? Its outputs are identified as Afrikaans (99.7%), with fewer unrecognised words than diffusion output (4.6%), though they

are less diverse (distinct-2 0.63 against 0.67 and 0.70). On length-standardised mGPT perplexity the large configuration is lower by 4.0 (95% CI [−7.5, −0.5]) and the small one higher by 7.7 ([3.7, 11.7]). The protocol's primary control, recombined slots from independent passages, produced only 2 outputs of ten words or fewer, too few for this comparison; the fitted-Gaussian contrasts were exploratory and were given this weight after the results were seen. A Gaussian prior over this latent space already yields fluent-scoring Afrikaans, and the larger denoiser improves on it only narrowly. For the named comparisons it agrees with mGPT in direction: the large-minus-small difference and the large configuration's advantage over AR are confirmed under both scorers (standardised ratios 0.84 and 0.74 under mGPT, 0.76 and 0.73 under Goldfish), but the small configuration's advantage over AR is not clear under either (mGPT 0.89 [0.76, 1.03]; Goldfish 0.95 [0.78, 1.16]). Two further checks failed their pre-set calibration and are not reported as results: MAUVE (Pillutla et al., 2021) with LaBSE embeddings (Feng et al., 2022) scored two halves of human text at 0.14 against a required 0.90, and two local model judges missed the validation thresholds for applying the proposal's sample checklist.

5.4 Interventions and full autoencoder training

Three decoder changes, each with a threshold fixed in advance, were tried on the 81 passages: decoding without the trigram ban, and retraining the last two or all twelve decoder blocks. None passed its threshold, but missing a threshold is not the absence of an effect. Retraining all twelve raised chrF from 67.70 to 70.05 (paired 95% CI for the change [0.78, 3.96]) and numerals from 21 to 25 of 33, a real but small gain, and stayed below the 70.70 chrF and 14-exact thresholds. Its best checkpoint lay at the end of its budget, so decoder-only adaptation had not been shown to saturate.

We then trained four new autoencoders under a protocol fixed before training (Table 5), all following the mT5-base recipe of Lovelace et al. (2023): every parameter trained, AdamW at learning rate 10^{-4}, effective batch 128, 500 warm-up updates and linear decay over 10,000, with one training seed per arm. The arms cross training data with latent length (32 or 64 slots). The extra data are 15,269 passages (511,155 words) of Gov-ZA cabinet statements (MIT licence) and the Vuk'uzenzele government newspaper (CC BY 4.0), processed like NCHLT, with duplicates of NCHLT passages removed and the 3 documents sharing a 12-word sequence with NCHLT validation or test dropped. Validation documents were divided into three disjoint panels: a 663-passage selection panel (whose documents also supply the 81 development passages), a 194-passage comparison panel and 103 reserve passages held back since 22 October. Arms were ranked on the selection panel, and the winner was evaluated once against the original on the reserve. At the original latent size the revised recipe raised exact matches on the comparison panel by 47.4 points (95% CI [40.4, 54.7]); longer latents added 4 to 7 points, and extra data at most 4. These are single-run contrasts. The winner met all three reserve thresholds: 73 of 103 exact against 19 (difference 52.4 points, CI [42.5, 63.3]), chrF 98.70 against 68.32, and numeral recall 1.00 against 0.77. Exact matches on long reserve passages remain rare (1 of 15), although their chrF is 95.9; whether the other 14 keep their meaning has not been checked by a reader. Diffusion on the new autoencoder was not complete for this version, so its effect on generation is not reported.

Arm	Data	Slots	Exact	chrF	Numerals
Original	NCHLT	32	26	69.24	0.62
A	NCHLT	32	118	96.35	0.94
B	NCHLT	64	126	97.89	1.00
C	+ extra	32	121	96.51	0.99
D	+ extra	64	134	98.20	1.00
Reserve, 103 passages					
Original	NCHLT	32	19	68.32	0.77
D (winner)	+ extra	64	73	98.70	1.00

Table 5: Autoencoder screen. Exact: exact reconstructions, out of 194 (comparison panel) or 103 (reserve). Numerals: recall of source numerals as a fraction, as in Table 2. + extra: NCHLT plus 511,155 words of other government-domain Afrikaans. Arms A–D train every parameter; the original trains only part of the model. Every arm stopped at its six-hour cap after about 3,500 of 10,000 planned updates.

5.5 A second language: Swahili

An earlier Swahili run of the same pipeline had been evaluated only by denoising loss and perplexity, the evaluation the Afrikaans checks showed to be insufficient, so we applied those checks to it without retraining. Its corpus is a supplied collection of 738 Swahili text files (23.6 million words)

whose original document boundaries could not be recovered, so it was split by file into 437,326 training, 29,761 validation and 25,537 test windows of at most 48 words; the files share some text across splits. The run used the same mT5 tokeniser (192-token cap, no truncation), two seeds per denoiser size and equal wall-clock budgets, so the small denoiser took about 172,000 updates and the large about 53,000; it has no AR baseline. Its test mGPT perplexity was 19.5 and 19.6 (small) and 20.6 and 20.8 (large), against 13.8 for human text. The checks followed a protocol approved in advance and used validation data only. Reconstruction of 200 passages reached mean chrF 74.8 (95% CI [72.9, 76.5]) against 24.3 with shuffled latents, but no exact matches, and numeral recall was 0.54. Random latents decoded by the same decoder scored clearly worse (perplexity 135) than diffusion samples. GlotLID labelled all diffusion samples and references as Swahili, against 57% of random-latent outputs. Diffusion outputs shared 10–11% of their word 4-grams with the training split, against 28% for references, which the supplied splits' cross-split overlap inflates. All readings are automatic and descriptive. One part of the Afrikaans pattern recurs: fluent-scoring samples coexist with an autoencoder that reconstructs no passage exactly, which perplexity alone would not have revealed. The other does not: Swahili random latents score far worse than diffusion samples, so there raw perplexity does separate the denoiser from the decoder.

6 Discussion

Verdict. *RQ1 is partly answered.* A reduced English control (ROCStories with BART, 5,000 updates) reached 93.26% validation token accuracy and 11 of 128 exact reconstructions, so the training code runs and learns; this does not show adequate fidelity, and the English and Afrikaans rates come from different models and data. Shorter runs showed that very small batches fail and that BF16 at effective batch 16 can diverge. It cannot rule out faults specific to the Afrikaans mT5 configuration. *RQ2's criterion is not met*: the original reconstructions change names, numerals and content words. The screen shows that much of this failure is avoidable with the same model and corpus: a revised recipe largely removes it at the original latent size, although surface similarity is not yet shown to be preserved meaning. *RQ3 is met only numerically.* The raw ratios are within 1.20 and distinct-2 shows no collapse, though decoding constraints partly prevent repetition; but the comparison is unmatched and the scorer length-sensitive, so under the pre-specified rule it is a weak diagnostic, and after length standardisation the small configuration's advantage over AR is not established. Fertility is reported descriptively (1.933 tokens per word). *The hypothesis is not supported*, because credibility required adequate reconstruction.

Why scores and fidelity diverge. Every sample passes through the same pretrained decoder with beam search, and both decoder and scorer were pretrained on large multilingual corpora that include Afrikaans (mT5 on mC4; mGPT on Wikipedia and C4). Random latents reach diffusion-level raw perplexity through this decoder, while reconstructions score 70.67 against 61.76 for their sources, so the decoder does not simply pull text towards high-probability Afrikaans. Raw perplexity therefore cannot separate what the denoiser contributes from what the decoder contributes, and it does not detect output that is not Afrikaans; the fitted-Gaussian control adds that much of what the scorers reward is already reachable by a simple prior over this latent space. Changing the denoiser alone does not repair the measured encode–decode mapping, but whether that mapping's errors limit unconditional generation, which has no source passage to preserve, is untested (compare Jiang et al., 2026). Our measurements also do not separate information missing from the latent from information the decoder fails to recover.

Under-adaptation or bottleneck? We call an autoencoder under-adapted when its trainable parameters have not been trained enough to use the latent it has; a bottleneck is a latent too small to hold the passage, and predicts errors that grow with length and gains from a longer latent. On the reserve passages the original reconstructs 16 of 47 short, 3 of 41 medium and 0 of 15 long passages exactly, against 45, 27 and 1 for arm D; arm D's long-passage chrF is 95.9, high surface similarity despite few exact matches; whether their content is preserved is unverified. At the original latent size the revised recipe added 47 points of exact match, and longer latents only 4 to 7, which argues against a fixed capacity limit as the main explanation. The recipe changes more than which parameters are trained: the batch size (128 against

eight), the schedule and the training history change too, and each arm is a single run. The evidence therefore favours a training problem over a bottleneck, but does not show which change matters; a matched comparison of frozen and fully trained autoencoders under one schedule, repeated over seeds, would.

7 Limitations and deviations

The study covers one administrative corpus, one language (Swahili is checked descriptively only) and one task, and training text was not screened with language identification. The diffusion system inherits mT5 pretraining and the AR baseline does not; they also differ in decoding and compute, and the AR budget was set provisionally against an earlier diffusion run, so differences cannot be attributed to diffusion versus autoregression. mGPT perplexity is length-dependent, chrF and numeral matching are surface measures, and low repetition is partly enforced by decoding. Seeds are conditional on one autoencoder and one AR run, each screen arm is a single training run, intervention gains are exploratory, and the test set is not untouched. Training rescales the self-conditioning estimate to norm eight per slot but sampling does not (Appendix A); this affects generation, not the reconstruction results, and its effect was not tested.

Several checks the evidence calls for remain to be done: blinded annotation of reconstruction errors by Afrikaans readers (entities, numbers, negation, predicate structure), including the 14 inexact long reserve passages; human calibration of the language-identification and dictionary labels, especially for short outputs; learning curves showing whether decoder-only adaptation had saturated; and generation from the retrained autoencoder under the same protocol. We make no claim about efficiency: end-to-end generation runs 250 denoising steps followed by beam decoding, and latency and memory were not measured.

Departures from the proposal: the English control used ROCStories with BART, the supported upstream route, as a reduced implementation check rather than the proposed OpenWebText reproduction; the tokeniser is mT5's rather than a new SentencePiece vocabulary; the AR baseline matches corpus and tokeniser only; and the planned researcher checklists for reconstructions and samples were not completed, with no reason documented. There was no participant-based evaluation, as planned; reconstruction and sample readings were AI-assisted and not verified by an Afrikaans reader. Additions not in the proposal are five seeds per configuration, the latent controls, the decoder interventions, the autoencoder screen and the Swahili checks.

8 Conclusion

Adapting latent diffusion to Afrikaans with limited in-domain data produced samples with lower raw perplexity than a small, historical autoregressive comparator, and lower again with a larger denoiser, from an autoencoder that does not reproduce its inputs faithfully. Random latents passed through the same decoder scored almost as well on raw perplexity, and latents drawn from a fitted Gaussian produced Afrikaans text that the larger denoiser beat only narrowly. Retraining the decoder alone gave a small gain; a revised full-training recipe substantially improved reconstruction on passages held back for that test. Whether better reconstruction yields better generation remains to be measured. In a new language, reconstruction fidelity, language identity and controls that bypass the denoiser should be reported with generation scores, because each can change what those scores mean.

Statements

AI use. AI assistance supported artifact auditing, analysis, sample and reconstruction inspection, drafting and LaTeX preparation; AI-assisted inspection is not human evaluation, and the author is responsible for the claims. **Ethics.** The corpus is publicly licensed and contact details were masked; no human participants were involved, and institutional ethics approval or exemption has not been verified. **Data and code.** Code, protocols and documentation: https://github.com/hadibutt3560/afrikaans-latent-diffusion; `REPRODUCIBILITY.md` at the repository root indexes the training configurations, scorer revisions, length-standardisation weights and band-support rule, and the interval procedures. Each study's folder holds its protocol, code and summary results, including per-run scores and the paired reconstructions of the 811-passage and train/held-out sets, with SHA-256 hashes of every output file. Model weights, corpora and full sample banks are not in the repository; the hashes allow regenerated outputs to be checked.

References

Ife Adebara, AbdelRahim Elmadany, Muhammad Abdul-Mageed, and Alcides Alcoba Inciarte. 2023. SERENGETI: Massively multilingual language models for Africa. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 1498–1537, Toronto, Canada. Association for Computational Linguistics.

Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. 2021. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems*, volume 34, pages 17981–17993.

Samuel R. Bowman, Luke Vilnis, Oriol Vinyals, Andrew M. Dai, Rafal Jozefowicz, and Samy Bengio. 2016. Generating sentences from a continuous space. In *Proceedings of the 20th SIGNLL Conference on Computational Natural Language Learning*, pages 10–21, Berlin, Germany. Association for Computational Linguistics.

Tyler A. Chang, Catherine Arnett, Zhuowen Tu, and Benjamin K. Bergen. 2024. Goldfish: Monolingual language models for 350 languages. *arXiv preprint arXiv:2408.10441*.

Ting Chen, Ruixiang Zhang, and Geoffrey Hinton. 2023. Analog bits: Generating discrete data using diffusion models with self-conditioning. In *International Conference on Learning Representations*.

Sander Dieleman, Laurent Sartran, Arman Roshannai, Nikolay Savinov, Yaroslav Ganin, Pierre H. Richemond, Arnaud Doucet, Robin Strudel, Chris Dyer, Conor Durkan, Curtis Hawthorne, Rémi Leblond, Will Grathwohl, and Jonas Adler. 2022. Continuous diffusion for categorical data. *arXiv preprint arXiv:2211.15089*.

Roald Eiselen and Martin J. Puttkammer. 2014. Developing text resources for ten South African languages. In *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC'14)*, pages 3698–3703, Reykjavik, Iceland. European Language Resources Association.

Fangxiaoyu Feng, Yinfei Yang, Daniel Cer, Naveen Arivazhagan, and Wei Wang. 2022. Language-agnostic BERT sentence embedding. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 878–891. Association for Computational Linguistics.

Alexey Gavrilov, Alan-Barsag Gazzaev, and Sergey Muravyov. 2026. Where quality breaks in compressed short-text generation: Staged bottleneck localization. *arXiv preprint arXiv:2607.24176*.

Shansan Gong, Mukai Li, Jiangtao Feng, Zhiyong Wu, and Lingpeng Kong. 2023. DiffuSeq: Sequence to sequence text generation with diffusion models. In *International Conference on Learning Representations*.

Ishaan Gulrajani and Tatsunori B. Hashimoto. 2023. Likelihood-based diffusion language models. In *Advances in Neural Information Processing Systems*, volume 36, pages 16693–16715.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, volume 33, pages 6840–6851.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. The curious case of neural text degeneration. In *International Conference on Learning Representations*.

Andrew Jaegle, Felix Gimeno, Andy Brock, Oriol Vinyals, Andrew Zisserman, and Joao Carreira. 2021. Perceiver: General perception with iterative attention. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 4651–4664.

Jiaxiu Jiang, Jingjing Ren, Wenbo Li, Bo Wang, Haoze Sun, Yijun Yang, Jianhui Liu, Yanbing Zhang, Shenghe Zheng, Yuan Zhang, Haoyang Huang, Nan Duan, and Wangmeng Zuo. 2026. TextLDM: Language modeling with continuous latent diffusion. *arXiv preprint arXiv:2605.07748*.

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. 2020. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 6282–6293, Online. Association for Computational Linguistics.

Amir Hossein Kargaran, Ayyoob Imani, François Yvon, and Hinrich Schütze. 2023. GlotLID: Language identification for low-resource languages. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 6155–6218. Association for Computational Linguistics.

Taku Kudo and John Richardson. 2018. SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 66–71, Brussels, Belgium. Association for Computational Linguistics.

Chunyuan Li, Xiang Gao, Yuan Li, Baolin Peng, Xiujun Li, Yizhe Zhang, and Jianfeng Gao. 2020. OPTIMUS: Organizing sentences via pre-trained modeling of a latent space. *arXiv preprint arXiv:2004.04092*.

Xiang Lisa Li, John Thickstun, Ishaan Gulrajani, Percy Liang, and Tatsunori B. Hashimoto. 2022. Diffusion-LM improves controllable text generation. In *Advances in Neural Information Processing Systems*, volume 35, pages 4328–4343.

Justin Lovelace, Varsha Kishore, Chao Wan, Eliot Shekhtman, and Kilian Q. Weinberger. 2023. Latent diffusion for language generation. In *Advances in Neural Information Processing Systems*, volume 36.

Clara Meister, Ryan Cotterell, and Tim Vieira. 2020. If beam search is the answer, what was the question? In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2173–2185. Association for Computational Linguistics.

Ivan Montero, Nikolaos Pappas, and Noah A. Smith. 2021. Sentence bottleneck autoencoders from transformer language models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 1822–1831. Association for Computational Linguistics.

Wilhelmina Nekoto, Vukosi Marivate, Tshinondiwa Matsila, Timi Fasubaa, Tajudeen Kolawole, et al. 2020. Participatory research for low-resourced machine translation: A case study in African languages. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 2144–2160, Online. Association for Computational Linguistics.

Alexander Quinn Nichol and Prafulla Dhariwal. 2021. Improved denoising diffusion probabilistic models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8162–8171.

Kelechi Ogueji, Yuxin Zhu, and Jimmy Lin. 2021. Small data? no problem! exploring the viability of pretrained multilingual language models for low-resourced languages. In *Proceedings of the 1st Workshop on Multilingual Representation Learning*, pages 116–126. Association for Computational Linguistics.

Krishna Pillutla, Swabha Swayamdipta, Rowan Zellers, John Thickstun, Sean Welleck, Yejin Choi, and Zaid Harchaoui. 2021. MAUVE: Measuring the gap between neural text and human text using divergence frontiers. In *Advances in Neural Information Processing Systems*, volume 34, pages 4816–4828.

Maja Popović. 2015. chrF: character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395. Association for Computational Linguistics.

Sello Ralethe. 2020. Adaptation of deep bidirectional transformers for Afrikaans language. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 2475–2478. European Language Resources Association.

Ehud Reiter. 2018. A structured review of the validity of BLEU. *Computational Linguistics*, 44(3):393–401.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695.

Tim Salimans and Jonathan Ho. 2022. Progressive distillation for fast sampling of diffusion models. In *International Conference on Learning Representations*.

Oleh Shliazhko, Alena Fenogenova, Maria Tikhonova, Anastasia Kozlova, Vladislav Mikhailov, and Tatiana Shavrina. 2024. mGPT: Few-shot learners go multilingual. *Transactions of the Association for Computational Linguistics*, 12:58–79.

Robin Strudel, Corentin Tallec, Florent Altché, Yilun Du, Yaroslav Ganin, Arthur Mensch, Will Grathwohl, Nikolay Savinov, Sander Dieleman, Laurent Sifre, and Rémi Leblond. 2022. Self-conditioned embedding diffusion for text generation. *arXiv preprint arXiv:2211.04236*.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. 2021. mT5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics.

Yizhe Zhang, Jiatao Gu, Zhuofeng Wu, Shuangfei Zhai, Joshua Susskind, and Navdeep Jaitly. 2023. PLANNER: Generating diversified paragraph via latent language diffusion model. In *Advances in Neural Information Processing Systems*, volume 36, pages 80178–80190.

A Model and estimators as implemented

Autoencoder. The mT5 encoder maps a passage of at most 128 tokens to hidden states, which the Perceiver compresses to $u \in \mathbb{R}^{32\times64}$. Each slot is rescaled to norm eight, $x_j = 8\,u_j/\|u_j\|$, so that x has unit variance per dimension on average, and a reconstruction map turns x into the decoder's cross-attention input. Training minimises token-weighted teacher-forced cross-entropy over the non-padding target tokens $y_1, \ldots, y_N$ of a batch, $\mathcal{L}_{\mathrm{AE}} = -\frac{1}{N}\sum_n \log p_\phi(y_n \mid y_{<n}, x)$.

Forward process. With $t \sim \mathcal{U}(0,1)$ and $\epsilon \sim \mathcal{N}(0, I)$, the noised latent is

$$z_t = \sqrt{\gamma(t)}\,x + \sqrt{1-\gamma(t)}\,\epsilon, \tag{1}$$

with $\gamma(t) = \cos^2(\pi t/2)$, where γ is the cumulative signal power ($\bar\alpha$ in DDPM notation), clipped so that $\log\frac{\gamma}{1-\gamma} \in [-15, 15]$. Latents are not shifted or scaled further.

Objective. The denoiser $f_\theta(z_t, t, c)$ predicts the velocity $v = \sqrt{\gamma}\,\epsilon - \sqrt{1-\gamma}\,x$, with unweighted squared error averaged over the 32×64 entries and the batch:

$$\mathcal{L} = \mathbb{E}_{x,t,\epsilon}\Big[\tfrac{1}{2048}\big\|f_\theta(z_t,t,c) - v\big\|_2^2\Big]. \tag{2}$$

Predictions convert as $\hat x_0 = \sqrt{\gamma}\,z_t - \sqrt{1-\gamma}\,\hat v$ and $\hat\epsilon = \sqrt{1-\gamma}\,z_t + \sqrt{\gamma}\,\hat v$.

Self-conditioning. With probability 0.5 the input c is a first-pass estimate, $c = \mathrm{sg}[\hat x_0(z_t, t, \varnothing)]$ with each slot rescaled to norm eight; otherwise $c = \varnothing$, a learned vector repeated over slots. During sampling, c is the previous step's $\hat x_0$ *without* that rescaling, so training and sampling present self-conditioning inputs of different norm. This is the untested discrepancy noted in §7.

Sampler. Sampling starts from $z_1 \sim \mathcal{N}(0, I)$ and takes 250 equal steps $t \to s$ with $s = t - 1/250$. With $\rho = \gamma(t)/\gamma(s)$ and fresh $\xi \sim \mathcal{N}(0, I)$,

$$z_s = \frac{1}{\sqrt{\rho}}\Big(z_t - \frac{1-\rho}{\sqrt{1-\gamma(t)}}\,\hat\epsilon\Big) + \sqrt{1-\rho}\,\xi, \tag{3}$$

and the last step returns $\hat x_0$, which is decoded without renormalisation. Checkpoints use exponential moving averages of the weights (decay 0.9999) and are selected on $\mathcal{L}$ estimated on 2,048 validation latents, three times with fixed seeds.

Scoring. Each output i is tokenised by the scorer into n_i tokens and scored after a start token, each token once, in 512-token windows with stride 256:

$$\mathrm{PPL} = \exp\Big(\frac{\sum_i \sum_{j=1}^{n_i} -\log p(w_{ij} \mid w_{i,<j})}{\sum_i n_i}\Big). \tag{4}$$

No end token is scored. For length-standardised perplexity, ℓ_b is the same token-weighted loss within word-length band b, and $\mathrm{PPL}_{\mathrm{std}} = \exp\big(\sum_b w_b\,\ell_b\big)$ with $w = (0.140, 0.223, 0.349, 0.288)$, the band shares of 1,000 human validation passages; it is reported only if every band holds at least 20 outputs. The large-minus-small permutation test is two-sided over all 252 divisions of the ten run means.

B Evaluation sets

Table 6 lists every set behind the reported numbers. Table 7 gives the values plotted in Figure 1.

Set	Size	Selection	Used for	First used
Training split	47,767 passages, 2,005 docs	Document split, seed 42	Training; overlap and copy checks	–
Audit passages	81 passages, 64 docs	Validation docs 0–63 of seed-91427 shuffle	Reconstruction, interventions	22 Oct.
Selection panel	663 passages	Same 64 documents	Screen: early stopping, ranking	27 Oct.
Comparison panel	194 passages	Validation docs 128–250	Screen contrasts; train/held-out check	27 Oct.
Reserve	103 passages	Validation docs 64–127	Screen winner vs original, once	31 Oct.
Human validation pool	4,825 passages	Non-reserve validation, $\leq$128 tokens	Language, dictionary, overlap	31 Oct.
Human scored subset	811 passages, 58 docs	1,000 scored validation rows minus reserve docs	Scorers; reconstruction of human text	30 Oct.
Diffusion validation	$5 \times 1{,}000$ per size	Sampling seeds 30000–30999	Controls, language, scorers	Before 23 Oct.
256-row subset	256 per run	First 256 seeds	AR ratios	22 Oct.
AR samples	256	Nucleus sampling	AR ratios, language	22 Oct.
Latent controls	1,000 per control	Seeds 30000+, 40000+, 50000+	Decoder-only controls	30 Oct.
Test banks	$5 \times 1{,}000$ per size; 1,000 human	Seeds 120000–120999	Test PPL, Figure 1	19 Sept. (AE), 9 Oct. (PPL)
Extra-data monitor	681 passages	5% of extra-data documents	Held out for a fresh evaluation	Not used here

Table 6: Evaluation sets. "First used" is when results on the set were first seen; the test banks were exposed before the models were frozen on 23 October.

Words	1–10	11–20	21–35	36+
Human (n)	240.8 (157)	54.1 (212)	32.6 (300)	25.9 (331)
Small, mean	332.1	89.4	53.6	40.2
range	294.7–386.9	82.4–94.5	52.2–55.0	39.1–40.8
outputs	797	851	1,419	1,933
Large, mean	213.1	69.3	47.4	38.1
range	188.0–246.1	63.7–73.3	46.4–48.7	37.6–39.3
outputs	739	1,262	1,579	1,420

Table 7: Test mGPT perplexity by output length (Figure 1). Diffusion: mean and range of five run values; outputs summed over runs. The range is not a confidence interval.